United International University

Department of Computer Science and Engineering

Final Term Examination || Trimester: Summer – 2025

School of Science and Engineering

Course: Math - 1151 (Fundamental Calculus)

Total marks – 40 || Duration – 2 hours

[*Note that the number of marks is given in brackets [ ] at the end of each question or part question. You are requested to answer all the questions in order*.]

Q1 **(a)** Given that $\frac{dy}{dx} = 6\sqrt{2x+1}$ and $y = 12$, when $x = 4$. Express/Find y in terms of x. **[3]**

(b) Find the equations of the normal to the curve $y = 1 + \cos 2x$ at $x = \frac{\pi}{2}$. **[4]**

(c) If $y = ln(2x^2+1)$, and $x = \sqrt{t^2+1}$, Express $\frac{dy}{dt}$ in terms of t. **[3]**

Total [10 marks]

Q2 **(a)** Find the coordinates of the turning point of the function $y = \frac{1}{3}x^3 - 2x^2 + 3x + 1$. Also, state their nature. **[4]**

(b) A delivery truck leaves a depot at time $t = 0$ seconds. Its velocity (in meters per second) for $0 \le t \le 5$ s is modeled by the function $v(t) = 3t^2 - 6t + 5$.

(i) Find the acceleration at $t = 2$. **[2]**

(ii) Find the displacement / position function $s(t)$. **[2]**

(iii) Find the time where the velocity is minimum. **[2]**

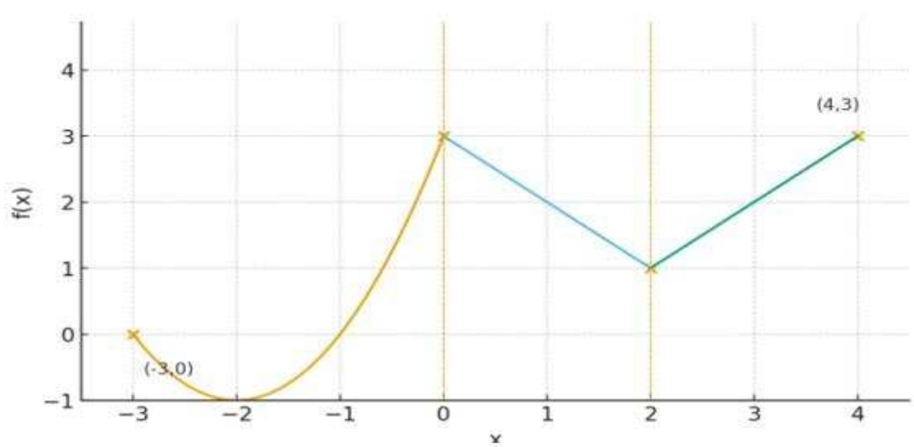
Total [10 marks]

Q3 **(a)** Differentiate the following w.r.t. x

(i) $y = (e^{-x}+1)^2$ **[3]**

(ii) $y = ln\sqrt[3]{(x+3)^2}$ **[3]**

(c) The function $y = f(x)$ is defined over the interval $-3 < x < 4$.



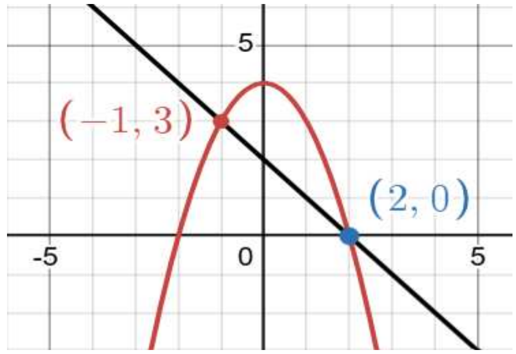
(i) State the values of x, where the function is not differentiable. **[2]**

(ii) Draw the derivative graph of $y = f(x)$. **[2]**

Total [10 marks]

[P.T.O]

Q4 **(a)** Find the area bounded by the curve $y = 4 - x^2$ and the line $y = 2 - x$. **[2]**



(b) Given that $f\,(x) = \begin{cases} 3x^2 + 1 & -2 \le x \le 1 \\ -2x + 6, & 1 < x \le 3 \end{cases}$. **[4]**

Sketch the graph of $f(x)$ and hence evaluate $\int_{-2}^{3} f(x)\ dx$.

(d) Evaluate the following integrals **(any two)** **[2+2]**

(i) $\int_0^1 x\sqrt{x^2+3}\ dx$

(ii) $\int_0^{\pi/4} (\cos 2x - \sin 2x)\ dx$

(iii) $\int_0^1 xe^{-x}\ dx$

Total [10 marks]